

Question ID: 7a5a74a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$3(2x - 6) - 11 = 4(x - 3) + 6$

If x is the solution to the equation above, what is the value of $x - 3$?

Answer

- A. $\frac{23}{2}$
- B. $\frac{17}{2}$
- C. $\frac{15}{2}$
- D. $-\frac{15}{2}$

Correct Answer: B

Rationale

Choice B is correct. Because 2 is a factor of both $2x$ and 6, the expression $2x - 6$ can be rewritten as $2(x - 3)$. Substituting $2(x - 3)$ for $(2x - 6)$ on the left-hand side of the given equation yields $3(2)(x - 3) - 11 = 4(x - 3) + 6$, or $6(x - 3) - 11 = 4(x - 3) + 6$. Subtracting $4(x - 3)$ from both sides of this equation yields $2(x - 3) - 11 = 6$. Adding 11 to both sides of this equation yields $2(x - 3) = 17$. Dividing both sides of this equation by 2 yields $x - 3 = \frac{17}{2}$.

Alternate approach: Distributing 3 to the quantity $(2x - 6)$ on the left-hand side of the given equation and distributing 4 to the quantity $(x - 3)$ on the right-hand side yields $6x - 18 - 11 = 4x - 12 + 6$, or $6x - 29 = 4x - 6$. Subtracting $4x$ from both sides of this equation yields $2x - 29 = -6$. Adding 29 to both sides of this equation yields $2x = 23$. Dividing both sides of this equation by 2 yields $x = \frac{23}{2}$.

Therefore, the value of $x - 3$ is $\frac{23}{2} - 3$, or $\frac{17}{2}$.

Choice A is incorrect. This is the value of x , not $x - 3$. Choices C and D are incorrect. If the value of $x - 3$ is $\frac{15}{2}$ or $-\frac{15}{2}$, it follows that the value of x is $\frac{21}{2}$ or $-\frac{9}{2}$, respectively. However, solving the given equation for x yields $x = \frac{23}{2}$. Therefore, the value of $x - 3$ can't be $\frac{15}{2}$ or $-\frac{15}{2}$.

